Information Geometry: The Fisher Metric

math-foundations / concept 40

First, an example

Let $X \sim \mathcal{N}(\mu, 1)$ with unknown mean $\mu \in \mathbb{R}$ and unit variance. The log-density is

$$\ell(\mu) \;=\; -\tfrac{1}{2}(x-\mu)^2 \;-\; \tfrac{1}{2}\log(2\pi).$$

Read aloud: ell of mu equals minus one-half times x minus mu squared, minus one-half log two pi. The score is $\partial_\mu \ell = x - \mu$, and $\mathbb{E}[x-\mu] = 0$ — the score has mean zero, always. The Fisher information is

$$F(\mu) \;=\; \mathbb{E}\big[(\partial_\mu \ell)^2\big] \;=\; \mathbb{E}[(x-\mu)^2] \;=\; 1.$$

Read aloud: F of mu equals expectation of partial-mu ell squared equals expectation of x minus mu squared equals one. Now compare two nearby Gaussians: a direct calculation gives

$$D_{\mathrm{KL}}(\mathcal{N}(0,1)\,\|\,\mathcal{N}(\epsilon,1)) \;=\; \tfrac{1}{2}\,\epsilon^2,$$

Read aloud: K-L divergence from N zero one to N epsilon one equals one-half epsilon squared. which is exactly $\frac{1}{2}\,F(\mu)\,\epsilon^2$. The Fisher metric *is* the second-order curvature of KL — that is the entire content of this lesson, in one example.

1 Setting

Let $\{p_\theta : \theta \in \Theta\}$ be a smoothly parameterized family of probability densities on a measurable space $\mathcal{X}$, with $\Theta \subseteq \mathbb{R}^d$ open. We assume sufficient regularity to interchange differentiation in θ with integration in x, and that $\log p_\theta(x)$ is twice continuously differentiable in θ for almost every x. Define the *score* $s_\theta(x) := \nabla_\theta \log p_\theta(x)$.

Definition 1 (Fisher information matrix)**.** *The Fisher information matrix at θ is the $d \times d$ matrix*

$$F(\theta) \;=\; \mathbb{E}_{X\sim p_\theta}\Big[s_\theta(X)\,s_\theta(X)^\top\Big] \;=\; -\,\mathbb{E}_{X\sim p_\theta}\big[\nabla_\theta^2 \log p_\theta(X)\big]\,.$$

Read aloud: *F of theta equals the expectation under X drawn from p-theta of the score times its transpose, which equals minus the expectation of the Hessian of log p-theta. The two expressions agree under the regularity assumptions above.*

Definition 2 (Statistical manifold)**.** *A* statistical manifold *is a pair (Θ, F) where Θ is a smooth manifold parameterizing $\{p_\theta\}$ and F is the Fisher information matrix viewed as a Riemannian metric tensor: for $u, v \in T_\theta\Theta$, $\langle u, v\rangle_\theta := u^\top F(\theta) v$.*

Definition 3 (Natural gradient)**.** *Given a smooth loss $L : \Theta \to \mathbb{R}$, the* natural gradient *is*

$$\widetilde{\nabla} L(\theta) \;=\; F(\theta)^{-1}\nabla L(\theta),$$

Read aloud: *natural-gradient L of theta equals F of theta inverse times the ordinary gradient of L at theta. defined wherever $F(\theta)$ is positive definite. It is the steepest-descent direction under the Fisher metric.*

2 The score has mean zero

Lemma 1. $\mathbb{E}_{X\sim p_\theta}[s_\theta(X)] = 0$.

Proof. Since $\int p_\theta(x)\,dx = 1$ for every θ, differentiating in θ and exchanging derivative and integral,

$$0 \;=\; \nabla_\theta \int p_\theta(x)\,dx \;=\; \int \nabla_\theta p_\theta(x)\,dx \;=\; \int \frac{\nabla_\theta p_\theta(x)}{p_\theta(x)}\,p_\theta(x)\,dx \;=\; \mathbb{E}_{p_\theta}[s_\theta(X)]. \qquad \square$$

Read aloud: zero equals the gradient in theta of the integral of p-theta of x dx, equals the integral of the gradient of p-theta dx, equals the integral of grad-p over p times p dx, equals the expectation of the score under p-theta.

3 Fisher is the local KL metric

Theorem 1. *The second-order Taylor expansion of $\epsilon \mapsto D_{\mathrm{KL}}(p_\theta \,\|\, p_{\theta+\epsilon})$ near $\epsilon = 0$ is*

$$D_{\mathrm{KL}}(p_\theta \,\|\, p_{\theta+\epsilon}) \;=\; \tfrac{1}{2}\,\epsilon^\top F(\theta)\,\epsilon \;+\; O\big(\|\epsilon\|^3\big)\,.$$

Read aloud: *K-L divergence from p-theta to p-theta-plus-epsilon equals one-half epsilon transpose F of theta epsilon, plus big-O of norm epsilon cubed.*

Proof. Define $\Phi(\epsilon) := D_{\mathrm{KL}}(p_\theta \,\|\, p_{\theta+\epsilon}) = \mathbb{E}_{X\sim p_\theta}[\log p_\theta(X) - \log p_{\theta+\epsilon}(X)]$. Set $\ell(\epsilon, x) := \log p_{\theta+\epsilon}(x)$, so $\Phi(\epsilon) = \mathbb{E}_{p_\theta}[\ell(0, X) - \ell(\epsilon, X)]$.

Taylor-expand $\ell(\epsilon, x)$ in ϵ at 0 to second order, with remainder $R(\epsilon, x) = O(\|\epsilon\|^3)$ uniformly on compact sets under our smoothness hypotheses:

$$\ell(\epsilon, x) \;=\; \ell(0, x) + \epsilon^\top \nabla_\epsilon \ell(0, x) + \tfrac{1}{2}\,\epsilon^\top \nabla_\epsilon^2 \ell(0, x)\,\epsilon + R(\epsilon, x).$$

Read aloud: ell of epsilon comma x equals ell at zero plus epsilon transpose times the gradient at zero plus one-half epsilon transpose Hessian at zero epsilon, plus a remainder R of epsilon x. Note $\nabla_\epsilon \ell(0, x) = s_\theta(x)$ and $\nabla_\epsilon^2 \ell(0, x) = \nabla_\theta^2 \log p_\theta(x)$. Substituting,

$$\ell(0, x) - \ell(\epsilon, x) \;=\; -\,\epsilon^\top s_\theta(x) - \tfrac{1}{2}\,\epsilon^\top \nabla_\theta^2 \log p_\theta(x)\,\epsilon - R(\epsilon, x).$$

Read aloud: ell at zero minus ell at epsilon equals minus epsilon transpose score, minus one-half epsilon transpose Hessian of log p-theta epsilon, minus the remainder. Take expectation under p_θ. By Lemma 1, $\mathbb{E}_{p_\theta}[\epsilon^\top s_\theta(X)] = \epsilon^\top \mathbb{E}_{p_\theta}[s_\theta(X)] = 0$, so the linear term vanishes. By the Hessian form of Fisher, $-\mathbb{E}_{p_\theta}[\nabla_\theta^2 \log p_\theta(X)] = F(\theta)$, so

$$\Phi(\epsilon) \;=\; 0 \;+\; \tfrac{1}{2}\,\epsilon^\top F(\theta)\,\epsilon \;-\; \mathbb{E}_{p_\theta}[R(\epsilon, X)] \;=\; \tfrac{1}{2}\,\epsilon^\top F(\theta)\,\epsilon \;+\; O\big(\|\epsilon\|^3\big)\,. \qquad \square$$

Read aloud: Phi of epsilon equals zero plus one-half epsilon transpose F of theta epsilon minus the expectation of the remainder, equals one-half epsilon transpose F of theta epsilon plus big-O of norm epsilon cubed.

Consequence. The Hessian of $\epsilon \mapsto D_{\mathrm{KL}}(p_\theta \| p_{\theta+\epsilon})$ at $\epsilon = 0$ is $F(\theta)$. The Fisher information is therefore the unique tensor that captures the local quadratic behaviour of KL divergence — justifying its role as the natural Riemannian metric on the statistical manifold and as the preconditioner in natural-gradient descent.